Method of expressing concentration of solution

61. 20 *g* of hydrogen is present in 5 *litre* vessel. The molar concentration of hydrogen is
(a) 4 (b) 1
(c) 3 (d) 2

62. To prepare a solution of concentration of 0.03 *g/ml* of $AgNO_3$, what amount of $AgNO_3$ should be added in 60 *ml* of solution
(a) 1.8
(b) 0.8
(c) 0.18
(d) None of these

63. How many grams of dibasic acid (mol. wt. 200) should be present in $100ml$ of its aqueous solution to give decinormal strength
(a) $1g$ (b) $2g$
(c) $10g$ (d) $20g$

64. The weight of pure $NaOH$ required to prepare $250cm^3$ of $0.1N$ solution is
(a) $4g$ (b) $1g$
(c) $2g$ (d) $10g$

65. If $20ml$ of $0.4NNaOH$ solution completely neutralises $40ml$ of a dibasic acid. The molarity of the acid solution is
(a) $0.1M$ (b) $0.2M$
(c) $0.3M$ (d) $0.4M$

66. Which of the following concentration factor is affected by change in temperature
(a) Molarity
(b) Molality
(c) Mole fraction
(d) Weight fraction

67. The distribution law is applied for the distribution of basic acid between
(a) Water and ethyl alcohol
(b) Water and amyl alcohol
(c) Water and sulphuric acid
(d) Water and liquor ammonia

68. Which is heaviest
(a) 25 gm of mercury
(b) 2 moles of water
(c) 2 moles of carbon dioxide
(d) 4 gm atoms of oxygen

69. The molarity of a solution of Na_2CO_3 having $10.6g/500ml$ of solution is
(a) $0.2M$ (b) $2M$
(c) $20M$ (d) $0.02M$

70. On passing H_2S gas through a solution of Cu^+ and Zn^{+2} ions, *CuS* is precipitated first because
(a) Solubility product of *CuS* is equal to the ionic product of *ZnS*

(b) Solubility product of *CuS* is equal to the solubility product of *ZnS*
(c) Solubility product of *CuS* is lower than the solubility product of *ZnS*
(d) Solubility product of *CuS* is greater than the solubility product of *ZnS*

71. The number of moles of solute per kg of a solvent is called its
(a) Molarity (b) Normality
(c) Molar fraction (d) Molality

72. 1.0 *gm* of pure calcium carbonate was found to require 50 *ml* of dilute HCl for complete reaction. The strength of the HCl solution is given by
(a) 4 *N* (b) 2 *N*
(c) 0.4 *N* (d) 0.2 *N*

73. Molecular weight of glucose is 180. A solution of glucose which contains 18 gms per litre is
(a) 2 molal (b) 1 molal
(c) 0.1 molal (d) 18 molal

74. 0.5 *M* of H_2SO_4 is diluted from 1 litre to 10 litre, normality of resulting solution is
(a) 1 *N* (b) 0.1 *N*
(c) 10 *N* (d) 11 *N*

75. If one mole of a substance is present in $1kg$ of solvent, then
(a) It shows molar concentration
(b) It shows molal concentration
(c) It shows normality
(d) It shows strength gm/gm

76. The molality of 90% H_2SO_4 solution is
[density=1.8 *gm/ml*]
(a) 1.8 (b) 48.4
(c) 9.18 (d) 94.6

77. The volume of water to be added to $100cm^3$ of 0.5 *N* H_2SO_4 to get decinormal concentration is
(a) 400 cm^3 (b) 500 cm^3
(c) 450 cm^3 (d) 100 cm^3

78. If 25 *ml* of 0.25 *M NaCl* solution is diluted with water to a volume of 500*ml* the new concentration of the solution is
(a) 0.167 *M* (b) 0.0125 *M*
(c) 0.833 *M* (d) 0.0167 *M*

79. 10 grams of a solute is dissolved in 90 grams of a solvent. Its mass percent in solution is
(a) 0.01 (b) 11.1
(c) 10 (d) 9

80. What is the molality of a solution which contains 18 *g* of glucose $(C_6H_{12}O_6)$ in 250 *g* of water

(a) 4.0 *m* (b) 0.4 *m*

(c) 4.2 *m* (d) 0.8 *m*

NEET IIT-JEE BOARD G10 EDUCATIONAL PLATFORM ESTD: 2005